```python
# Problem 1 – Student Report Card

students = { "Rahul": 91,  "Ayesha": 84,"Kiran": 67, "Ali": 45}

passed = 0

failed = 0

for name, marks in students.items():

    print(name, ":", marks)

    if marks >= 40:

        print("Pass")

        passed += 1

    else:

        print("Fail")

        failed += 1

print("Passed:", passed)

print("Failed:", failed)


# Problem 2 – Grocery Store Inventory

inventory = {"Rice": 20,"Sugar": 15,"Milk": 8, "Eggs": 30}

item = "Milk"

if item in inventory:

    inventory[item] -= 1

else:

    print("Item Not Available")

print(inventory)


# Problem 3 – Login System

users = { "admin": "python123", "student": "welcome","guest": "guest123"}

attempt = 1
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while attempt <= 3:
    username = input("Username: ")
    password = input("Password: ")
    if username in users and users[username] == password:
        print("Login Successful")
        break
    else:
        print("Invalid Login")
        attempt += 1
if attempt > 3:
    print("Account Locked")

# Problem 4 – Library Book Search
books = {101: "Python",102: "Java", 103: "SQL", 104: "Machine Learning"}
try:
    book_id = int(input("Enter Book ID: "))
    if book_id in books:
        print(books[book_id])
    else:
        print("Book Not Found")
except:
    print("Invalid Input")

# Problem 5 – Bank Balance Checker
customers = { "Rahul": 25000, "Ayesha": 30000, "Ali": 18000}
name = input("Enter customer name: ")
if name in customers:
    print("Balance:", customers[name])
```

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else:
    print("Customer Not Found")

# Problem 6 – Daily Expense Tracker
expenses = {}
while True:
    expense = input("Expense Name (stop to end): ")
    if expense.lower() == "stop":
        break
    amount = float(input("Amount: "))
    expenses[expense] = amount
print(expenses)
print("Total Expenses:", len(expenses))

# Problem 7 – Student Attendance File
with open("attendance.txt", "w") as file:
    file.write("Rahul\n")
    file.write("Ayesha\n")
    file.write("Ali\n")
    file.write("Kiran\n")
with open("attendance.txt", "r") as file:
    print(file.read()

# Problem 8 – Personal Diary
text = input("Write today's experience: ")
with open("diary.txt", "w") as file:
    file.write(text)
```

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print("Diary Saved Successfully")


# Problem 9 – Read Missing File
try:
   with open("marks.txt", "r") as file:
      print(file.read())
except FileNotFoundError:
   print("File Not Found")


# Problem 10 – Safe Calculator
try:
   a = float(input("Enter first number: "))
   b = float(input("Enter second number: "))
   print("Result:", a / b)
except ValueError:
   print("Invalid Number")
except ZeroDivisionError:
   print("Cannot Divide by Zero")


# Problem 11 – High Score Board
scores = { "Rahul": 120,"Ali": 250,"John": 180}
top_player = ""
highest = 0
for player, score in scores.items():
   if score > highest:
      highest = score
      top_player = player
```

```
print(top_player, highest)

# Problem 12 – Hospital Patient Records
patients = {101: "Rahul",102: "Sita",103: "Kiran",104: "Ali"}
try:
  pid = int(input("Enter Patient ID: "))
  if pid in patients:
    print(patients[pid])
  else:
    print("Invalid ID")
except:
  print("Invalid Input")

# Problem 13 – Movie Ticket Booking
seats = { "A1": "Available", "A2": "Booked", "A3": "Available", "A4": "Available"}
seat = input("Enter Seat Number: ")
if seat in seats:
  if seats[seat] == "Available":
    seats[seat] = "Booked"
    print("Seat Booked")
  else:
    print("Seat Already Booked")
else:
  print("Invalid Seat")
print(seats)
```

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# Problem 14 – Weather Logger
temp = input("Enter Today's Temperature: ")
with open("weather.txt", "a") as file:
   file.write(temp + "\n")
print("Temperature Saved")


# Problem 15 – Shopping Bill Generator
items = {"Milk": 50,"Bread": 40, "Eggs": 70,"Rice": 500}
bill = 0
while True:
   item = input("Enter Item (done to finish): ")
   if item.lower() == "done":
      break
   if item in items:
      bill += items[item]
   else:
      print("Item Not Found")
print("Final Bill:", bill)


# Problem 16 – Cab Booking System
fare = {"Hyderabad": 300, "Warangal": 700,"Karimnagar": 600}
city = input("Enter Destination: ")
if city in fare:
   print("Fare:", fare[city])
else:
   print("Service Not Available")
```

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# Problem 17 – Password Strength Checker
while True:
    password = input("Create Password: ")
    if len(password) >= 8:
        print("Password Accepted")
        break
    else:
        print("Password Too Short")

# Problem 18 – Word Counter
with open("sample.txt", "w") as file:
    file.write("Python is easy to learn.")
with open("sample.txt", "r") as file:
    data = file.read()
    words = data.split()
print("Total Words:", len(words))

# Problem 19 – Student Database
student = { "Name": "Rahul","Age": 20,"Course": "Python",  "City": "Hyderabad"}
print(student)
print(student.keys())
print(student.values())
print(student.items())
student.update({"City": "Warangal"})
print(student)
```

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# Problem 20 – Space Mission Log
systems = {"Engine": "OK","Fuel": "OK","Navigation": "FAIL","Communication": "OK"}
failed = False
for system, status in systems.items():
   if status == "FAIL":
      with open("mission_log.txt", "w") as file:
         file.write(system + " FAILED")
      print(system)
      print("Mission Aborted")
      failed = True
      break
if not failed:
   with open("mission_log.txt", "w") as file:
      file.write("Launch Successful")
   print("Launch Successful")
```